

# ANSH TOMAR

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## EDUCATION

**University of Illinois Urbana-Champaign**

**December 2025**

*Bachelor of Science in Computer Science and Statistics*

**GPA: 3.9/4.0**

- **Coursework:** Data Structures, Algorithms, Applied Machine Learning, System Programming, Computer Architecture, Discrete Structures, Linear Algebra, Calculus III, Numerical Methods, Programming for Information Problems, Software Design Lab, Data Science Discovery, Data Science Exploration, Statistical Modeling II, Statistics and Probability II, Fundamental Mathematics
- **Awards:** Dean's List (Fall 2022, Spring 2023)

## EXPERIENCE

**UL Solutions**

**May 2025 – August 2025**

*Data Analyst & Machine Learning Intern*

*Northbrook, Illinois*

- Built predictive models (Random Forest, Gradient Boosting) in Python (scikit-learn, pandas, NumPy) to estimate material degradation with. ~ 75% accuracy
- Processed and organized 10,000+ data samples through data cleaning, normalization, and outlier detection
- Performed exploratory data analysis and feature engineering to identify high-importance variables and improve model performance
- Implemented train/test/holdout validation for robust performance evaluation and reproducibility

**James River Insurance Company**

**May 2023 – August 2024**

*Data & Cloud Engineer Intern*

*Richmond, Virginia*

- Led and developed a prototype involving Azure Cognitive Search for the purposes of optimizing dataset retrieval and fraud detection for thousands of insurance claim documents and images and displayed findings towards senior executives
- Created multiple pricing models for applications of different Microsoft services in various projects & determining future procedures within projects
- Worked on multiple SQL scripts to provide reports of 1000+ claims data through filtering out unnecessary data, converting data into readable tables exported through Excel, and optimizing scripts to dynamically produce data for each preceding and passing month
- Edited and ran VBA scripts to update Microsoft Excel macros for 1000+ insurance spreadsheets to update data to latest versions of Excel functionality
- Developed and executed SQL scripts to perform data migrations 7 years' worth of claims data from Excel to cloud through Azure Data Factory pipelines

**Department of Statistics**

**January 2025 – May 2025**

*Undergraduate Teaching Assistant – Statistics and Probability II*

*Champaign, Illinois*

- Led discussions and office hours for 100+ students on advanced statistical topics, including moment-generating functions, maximum likelihood estimation, and hypothesis testing
- Graded assignments and exams, providing timely and constructive feedback for student learning and improvement
- Collaborated with instructors and other teaching assistants to address student inquiries and facilitate learning, such as by using R to showcase statistical methods done during the course

**Illinois Data Science Club**

**September 2024 – December 2024**

*Data Analyst Project Member*

*Champaign, Illinois*

- Conducted data analysis on NBA player injury rates, employing techniques such as data cleaning and linear regression using Python's Pandas library to explore correlations with in-game statistics
- Developed informative visualizations to identify and present data trends using Python libraries like Matplotlib and Seaborn

### **Digsfact**

*Machine Learning Intern*

**January 2023 – July 2023**

*Naperville, Illinois*

- Applied machine learning techniques such as supervised learning and Python to understand company operations done with training and predictive models

### **Code Ninjas**

*Tutor*

**July 2021 – September 2022**

*Naperville, Illinois*

- Instructed over 50 students in introductory and intermediate game development using platforms such as Scratch, Roblox, and JavaScript
- Guided onboarding sessions for 3 new students, introducing them to game design principles through hands-on Scratch projects
- Developed and managed discussion topics, STEM activities, and learning materials to create an engaging and productive learning environment

## **PROJECTS**

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### **Music Player**

*Java*

- Developed a linked-queue-based music player with playlist management and song filtering
- Implemented file handling for loading playlists and adding songs
- Integrated audio playback controls through a keyboard-driven user menu
- Precise unit testing performed for validation of every implementation

### **Prairie Learn Enhancements**

*HTML / Typescript*

- Developed a Chrome extension to enhance the Prairie Learn platform utilized by majority of UIUC students by displaying convenient assignment details such as upcoming deadlines, completion progress, and assignment descriptions to improve student task/time management

### **Time Zone Converter**

*Python / Tkinter*

- Designed and developed a GUI-based time zone converter of 20+ regions around the world using Python
- Utilized Tkinter to apply personal designs of an interactive user interface with dropdown menus for time zone selection including a structured layout and color customization

### **Meal Planner**

*Java*

- Designed a Java-based application to manage and organize food item data
- Implemented functionality to add, remove, search, and list food items, with user-friendly prompts for calorie information
- Enabled loading of pre-existing food lists or manual input for seamless user experience

## **CERTIFICATIONS**

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## **Data Science DISCOVERY**

**May 2024**

*University of Illinois at Urbana Champaign*

- Proficient in data analysis, probability, and machine learning using Python, with expertise in statistical inference, predictive modeling, and Git for version control

## **Python Essential Training**

**July 2024**

*LinkedIn Learning*

- Developed a strong foundation in Intermediate Python programming, including data structures, functions, file handling, and error handling
- Applied Python to real-world scenarios, including data manipulation and automation

## **TECHNICAL SKILLS**

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- **Programming Languages:** Java, Python, C, SQL, R, JavaScript, HTML/CSS
- **Libraries/Frameworks:** Pandas, NumPy, Scikit-Learn, BeautifulSoup, Matplotlib, Seaborn, SciPy, Tkinter
- **Tools/Software:** Git, Microsoft Azure (Cognitive Services, Pipeline, Data Lake Storage, DevOps), Microsoft SQL Server Management Studio, Adobe Creative Cloud, Visual Studio, Eclipse, IntelliJ, Jupyter Notebook, PyCharm, Visual Basic for Applications, Microsoft 365